

GIGBOX GPIO WIRING PLAN

Raspberry Pi 5 | BCM numbering | active-low inputs

SAFETY

- Connect the other side of every control to GND.
- Use internal pull-ups; never apply 5 V to GPIO.
- GPIO 18, 19, 20, 21 are reserved for I2S audio.
- Power off the Pi before wiring and verify with a meter.

MAIN ENCODER

Signal	BCM	Physical pin
Channel A	5	29
Channel B	6	31
Push switch	13	33
Common	GND	any ground pin

FIVE-WAY NAVIGATION

Signal	BCM	Physical pin
UP	17	11
DOWN	27	13
LEFT	22	15
RIGHT	23	16
CLICK	24	18
Common	GND	any ground pin

GROUND PINS

Physical pins 6, 9, 14, 20, 25, 30, 34, or 39.

All inputs use internal pull-ups and press when connected to ground.

GIGBOX BUTTON MAP

Button	Function	BCM	Physical pin
1	Transpose up (+1 semitone)	16	36
2	Transpose down (-1 semitone)	7	26
3	Octave up (+12 semitones)	8	24
4	Octave down (-12 semitones)	9	21
5	Sustain on/off, MIDI CC 64	25	22
6	Play/pause	26	37
7	Main menu	12	32
8	Mixer	4	7
9	ZS3	10	19
10	ALT	11	23

AUDIO RESERVATION - DO NOT USE FOR CONTROLS

I2S clock BCM 18 Physical pin 12
I2S frame sync BCM 19 Physical pin 35
I2S data in BCM 20 Physical pin 38
I2S data out BCM 21 Physical pin 40

BRING-UP CHECKLIST

- ☐ Pi powered off during wiring
- ☐ All control commons connected to GND
- ☐ No control wire connected to GPIO 18-21
- ☐ Encoder A/B are not swapped
- ☐ Every input reads released before pressing
- ☐ Test one control at a time with Zynthian control test

See GIGBOX_GPIO_WIRING_PLAN.md for the complete reference.